Mansoura University
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MODELLING & SIMULATION

SW,BI,AI

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COURSE SYLLABUS

- Chapter 1: Introduction.
- Chapter 2: Probability as Using in Simulation.
- Chapter 3: Queueing Simulation.
- Chapter 4: Inventory Simulation.
- Chapter 5: Random-Number Generation.
- Chapter 6: Input Modeling.
- Chapter 7: Output Analysis.

COURSE SYLLABUS

- Chapter 1: Introduction.
- Chapter 2: Probability as Using in Simulation.
- Chapter 3: Queueing Simulation (Part 1)
- Chapter 4: Inventory Simulation.
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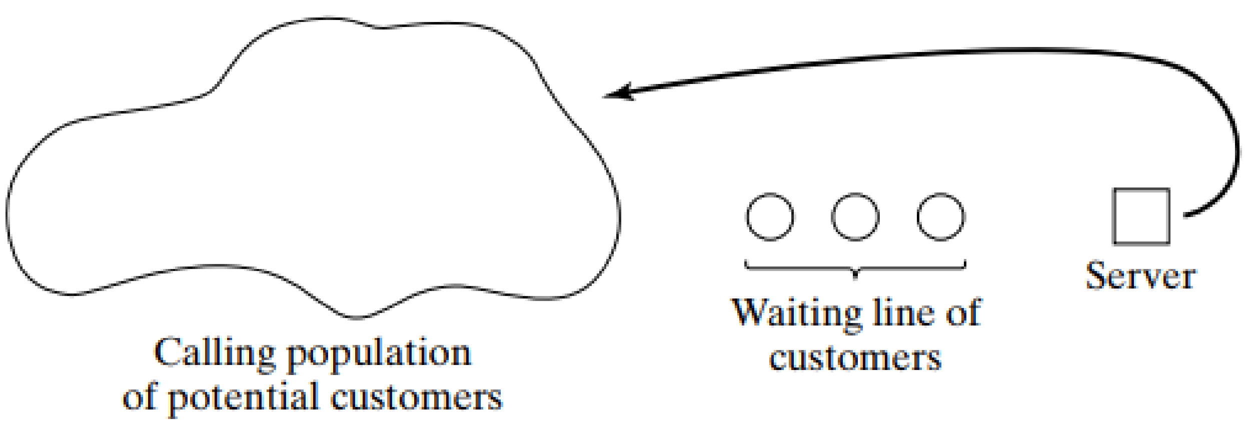
CHAPTER CONTENT: QUEUEING SIMULATION

- Introduction.
- Description of Queuing System.
- Simulating a Single-Server Queue.
  - Example.
- Simulating a Queue with Multi-Servers.
  - Example.

INTRODUCTION

- **Queueing Model:**
- Queueing models are among the most widely used applications of simulation in practice.
- Simulation is often used in the analysis of queueing models.
- In a simple but typical queueing model, customers arrive from time to time and join a queue (waiting line), are eventually served, and finally leave the system.



INTRODUCTION

- Whether we are waiting in line at a bank, a grocery store, a call center, or a traffic light, we encounter queueing systems daily.
- Simulation provides a powerful tool for analyzing these systems, understanding their behavior, and designing improvements.

WHY SIMULATE QUEUEING SYSTEMS?

- Queueing models, whether solved mathematically or analyzed through simulation, provide analysts with a powerful tool for designing and evaluating the performance of queueing systems.
- The insights gained help in:
  - Designing new facilities: Determining how many servers are needed
  - Improving existing systems: Identifying bottlenecks and reducing wait times
  - Balancing tradeoffs: Making decisions that balance server utilization against customer satisfaction

THE TRADEOFF PROBLEM

- When designing or improving a queueing system, the analyst (or decision maker) is involved in tradeoffs between server utilization and customer satisfaction in terms of line lengths and delays.
  - Adding more servers reduces waiting times but increases costs
  - Reducing servers saves money but may lead to long queues and dissatisfied customers
  - Simulation helps find the optimal balance point

TYPICAL MEASURES OF PERFORMANCE

- When analyzing queueing systems, we typically focus on several key performance measures:

Performance Measure	Description
Server Utilization	Percentage of time a server is busy
Queue Length	Number of customers waiting in line
Customer Delays	Time customers spend waiting before service
System Time	Total time customers spend in the system (waiting + service)
Probability of Waiting	Likelihood that an arriving customer must wait

DESCRIPTION OF QUEUING SYSTEM

- The key elements of a queueing system are the customers and servers.

Element	Definition	Examples
Customers	Anything requiring service	People, machines, trucks, patients, airplanes, e-mail
Servers	Resources providing service	Receptionists, repair personnel, medical staff, runways

DESCRIPTION OF QUEUING SYSTEM

- **Queueing System:**

  1) The Calling Population.
  2) System Capacity.
  3) The Arrival Process.
  4) Queue Behavior and Queue Discipline.
  5) Service Times and the Service Mechanism.

THE CALLING POPULATION

- Also called: Input source (source of customers).
- The population of potential customers, referred to as the calling population, may be assumed to be finite or infinite.
- Infinite size:
  - Simulation assumptions: If a unit leaves the calling population and joins the waiting line or enters service, there is no change in the arrival rate (the average number of arrivals per unit of time) of other units that may need service.
  - Most simulated models.
- Finite size:
  - Less common and complicates the simulation.

THE CALLING POPULATION

- The main difference between finite and infinite population models is how the arrival rate is defined.
- In an infinite population model, the arrival rate (i.e., the average number of arrivals per unit of time) is not affected by the number of customers who have left the calling population and joined the queueing system.

SYSTEM CAPACITY:

- System capacity is the maximum number of units that can be accommodated in the system (waiting line + service).
  - Limited Capacity: There is a physical limit to how many customers can wait (e.g., a small waiting room)
  - Unlimited Capacity: The system can accommodate any number of units in the waiting line (common assumption in many models)
- When system capacity is limited, arriving customers may be turned away (blocked) if the system is full.

THE ARRIVAL PROCESS:

- The arrival process for infinite-population models is usually characterized in terms of interarrival times of successive customers.

Arrival Type	Description	Example
Scheduled Arrivals	Customers arrive at predetermined times	Appointments at a doctor's office
Random Arrivals	Arrivals occur at unpredictable times	Customers at a grocery store

When arrivals occur at random times, the interarrival times are usually characterized by a **probability distribution**. Common distributions include:

- Exponential distribution (memoryless property)
- Uniform distribution (all values equally likely)
- Normal distribution (for scheduled arrivals with variation)

QUEUE BEHAVIOR AND QUEUE DISCIPLINE

- **Queue behavior** refers to the actions of customers while in a queue waiting for service to begin.

Balk	Leave upon arrival when they see that the line is too long
Reneg	Leave after being in the line when they see that the line is moving too slowly
Jockey	Move from one line to another if they think they have chosen a slow line

QUEUE BEHAVIOR AND QUEUE DISCIPLINE

- **Queue discipline** refers to the logical ordering of customers in a queue and determines which customer will be chosen for service when a server becomes free.

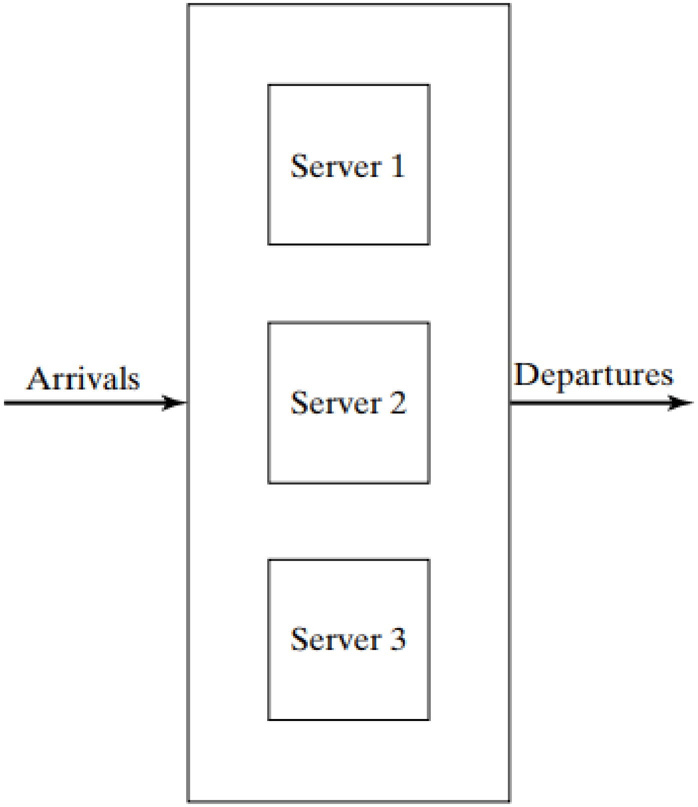
FIFO (First-In-First-Out)	Customers served in order of arrival; also called FCFS (First-Come-First-Served)
LIFO (Last-In-First-Out)	Most recent arrival served first; also called LCFS
SIRO (Service In Random Order)	Customer selected randomly from the queue
SPT (Shortest Processing Time First)	Customer with shortest expected service time served first
PR (Priority)	Customers served according to priority classes

SERVICE TIMES AND THE SERVICE MECHANISM

- The service times of successive arrivals are denoted by $S_1, S_2, S_3, \ldots$ They may be:
  - Constant: Every customer takes exactly the same service time
  - Random: Service times vary according to a probability distribution
- In the random case, $\{S_1, S_2, S_3, \ldots\}$ is usually characterized as a sequence of independent and identically distributed (IID) random variables.
- In some systems, service times may depend upon:
  - Time of day: Servers work faster during busy periods
  - Length of the waiting line: Servers may speed up when lines are long

SERVICE TIMES AND THE SERVICE MECHANISM

- A queueing system consists of a number of service centers and interconnecting queues.
- Each service center consists of some number of servers, c, working in parallel; that is, upon getting to the head of the line, a customer takes the first available server.



DESCRIPTION OF QUEUING SYSTEM

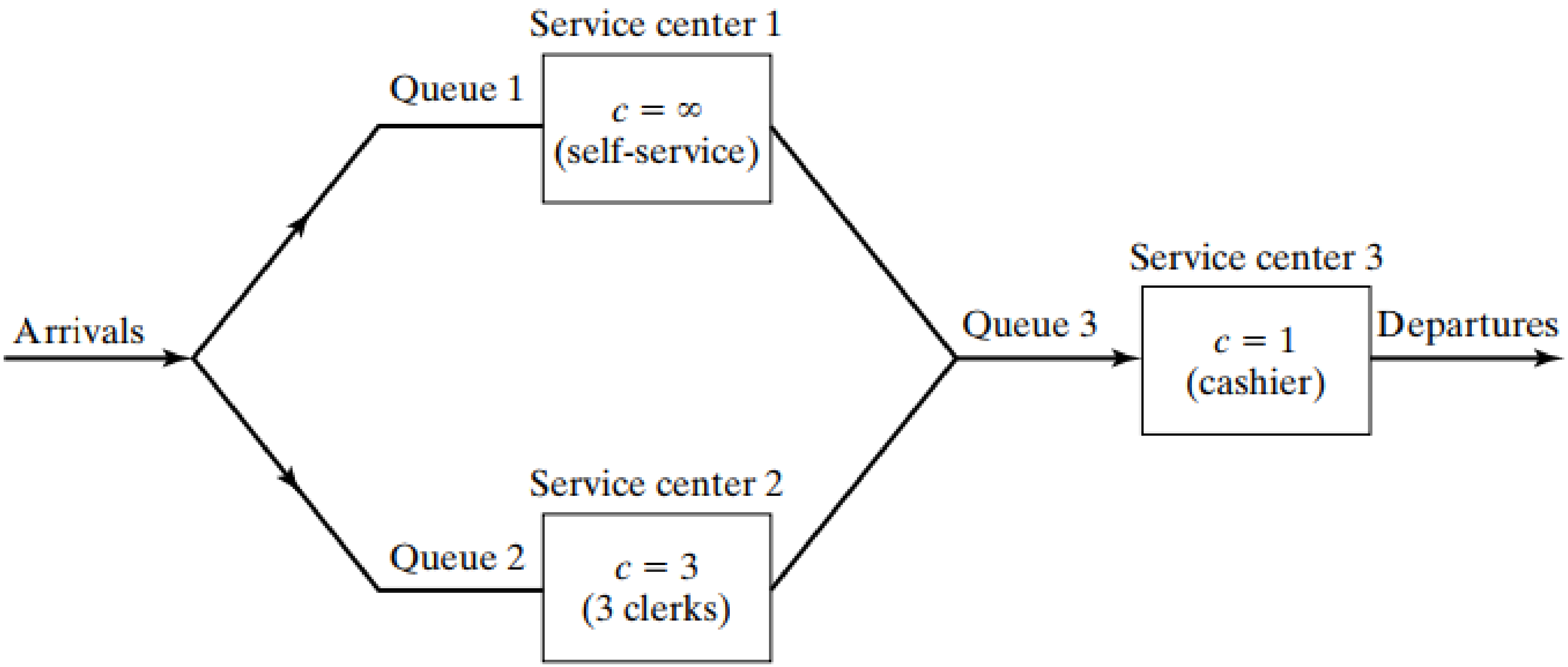
- A notational system for parallel server systems: A/B/c/N/K.
- These letters represent the following system characteristics:

Symbol	Meaning	Common Examples
A	Arrival time distribution	M (Exponential), D (Deterministic), G (General)
B	Service time distribution	M (Exponential), D (Deterministic), G (General)
c	Number of parallel servers	1, 2, 3, ...
N	System capacity (optional)	If omitted, assumed infinite
K	Size of calling population (optional)	If omitted, assumed infinite

DESCRIPTION OF QUEUING SYSTEM

- **Examples:**
- M/M/1: Exponential interarrivals, exponential service times, single server
- M/D/2: Exponential interarrivals, deterministic service times, two servers
- G/G/∞: General interarrivals, general service times, infinite servers

SERVICE TIMES AND THE SERVICE MECHANISM

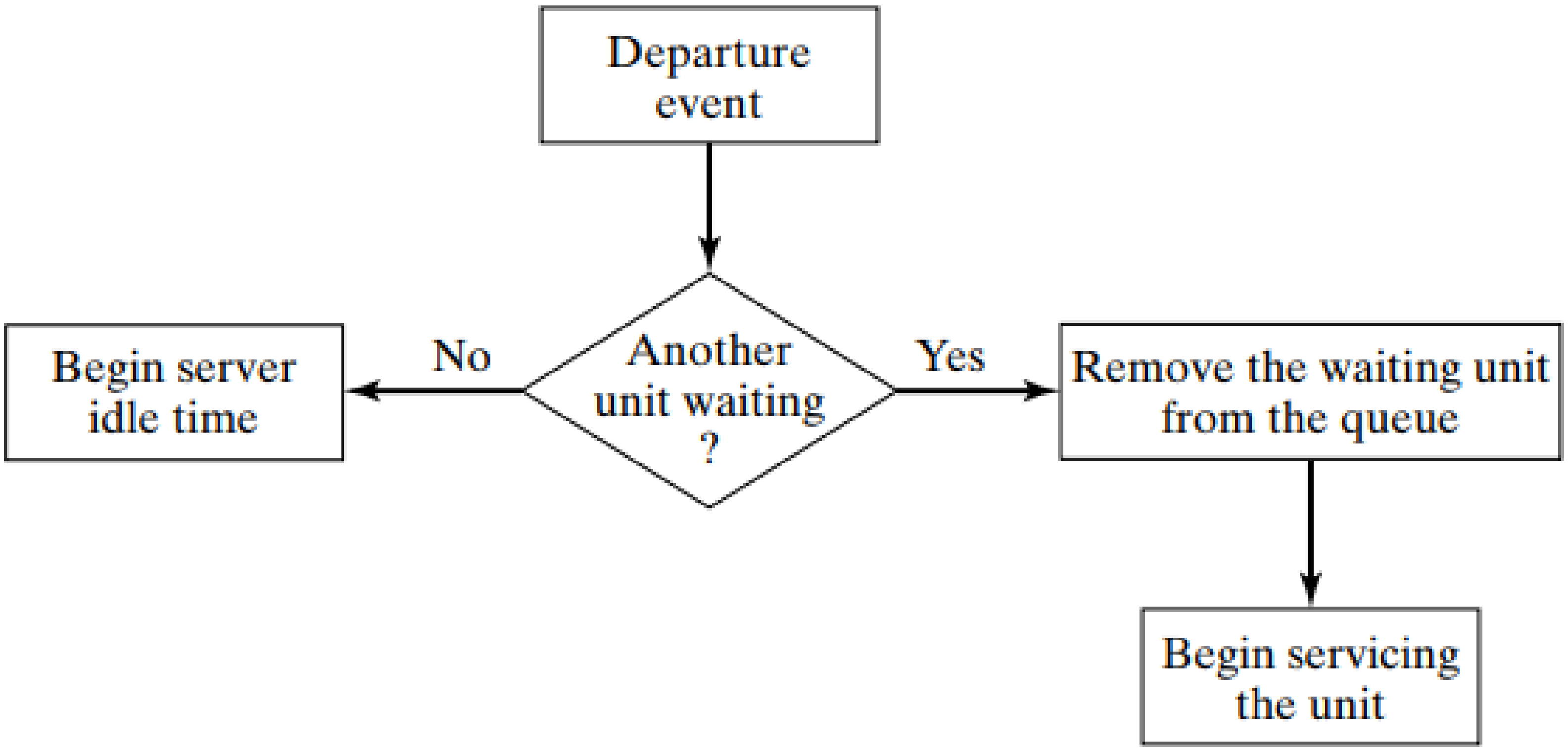


QUEUING SYSTEM STATE

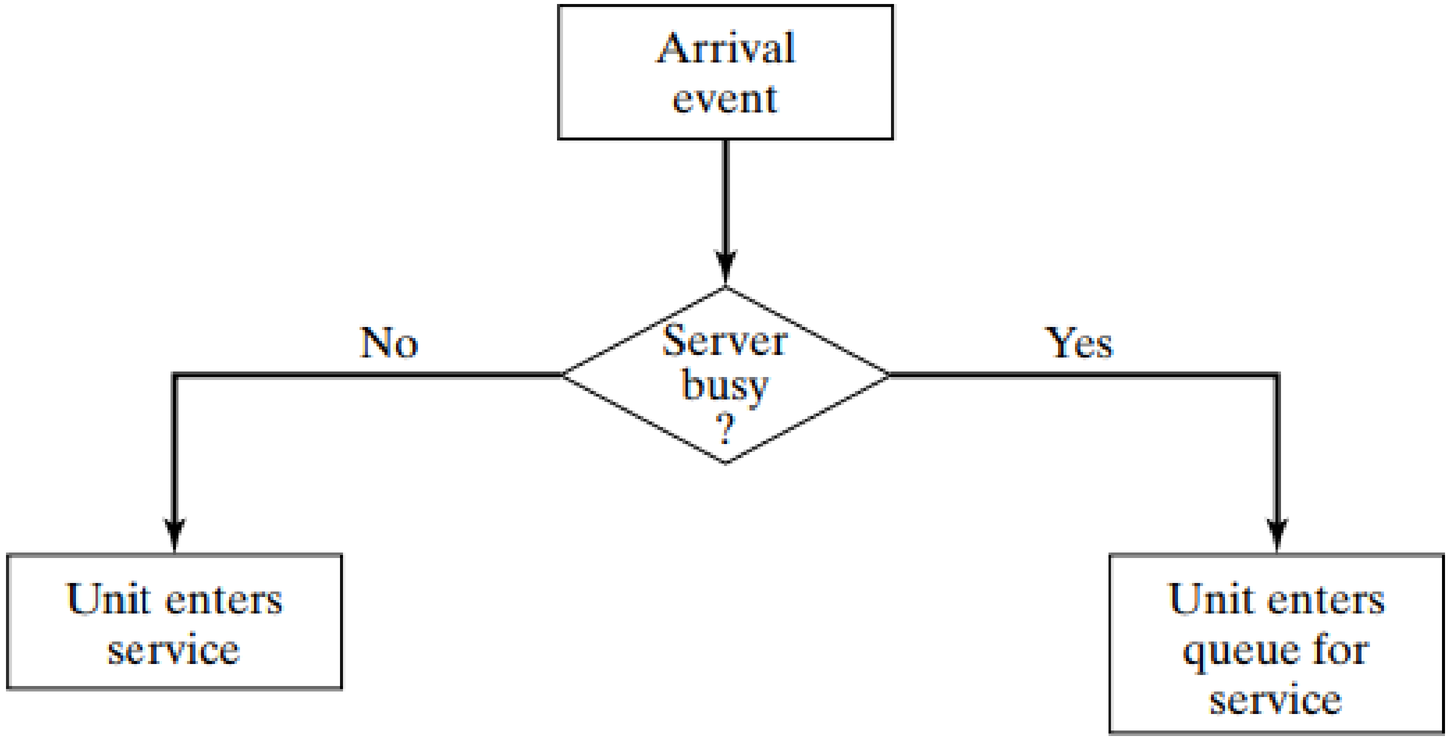
Simulating a Single-Server Queue

- To simulate a queueing system, we must track the **state** of the system—the quantities that collectively completely describe the system at any given instant from the viewpoint of the study objectives.
  - Number of units in the system: How many units are currently in the queue or being processed by the server.
  - Status of server (idle, busy): Is the server(s) busy with processing some units or idle waiting for a job.

QUEUING SYSTEM STATE

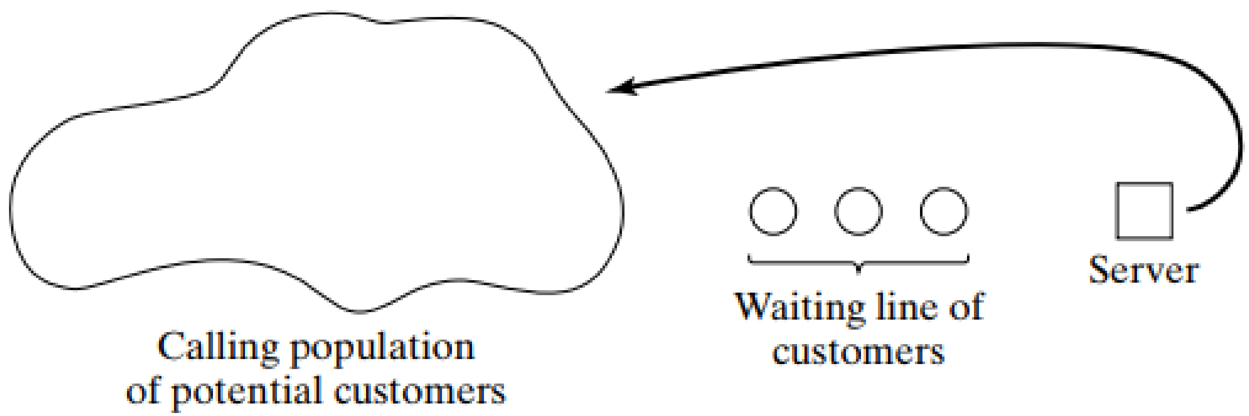


QUEUING SYSTEM STATE



SIMULATION TABLE

- The following table was designed as a simulation table specifically for a single-channel queue that serves customers on a first-in–first-out (FIFO) basis.



KEY CALCULATIONS

- **For each customer, we compute:**

1. Arrival Time: Arrival Time of previous customer + Time Between Arrivals
2. Time Service Begins: max(Arrival Time, Time Service Ends of previous customer)
3. Time Service Ends: Time Service Begins + Service Time
4. Waiting Time in Queue: Time Service Begins - Arrival Time
5. Time Spent in System: Time Service Ends - Arrival Time (or Waiting Time + Service Time)
6. Server Idle Time: Time Service Begins - Time Service Ends of previous customer (if positive)

SIMULATION TABLE

The interarrival and service times are taken from distributions!

Customer	Interarrival Time	Arrival Time on Clock	Service Time
1	-	0	2
2	2	2	1
3	4	6	3
4	1	7	2
5	2	9	1
6	6	15	4

The simulation run is build by meshing clock, arrival, and service times!

Customer Number	Arrival Time [Clock]	Time Service Begins [Clock]	Service Time [Duration]	Time Service Ends [Clock]
1	0	0	2	2
2	2	2	1	3
3	6	6	3	9
4	7	9	2	11
5	9	11	1	12
6	15	15	4	19

SIMULATION TABLE

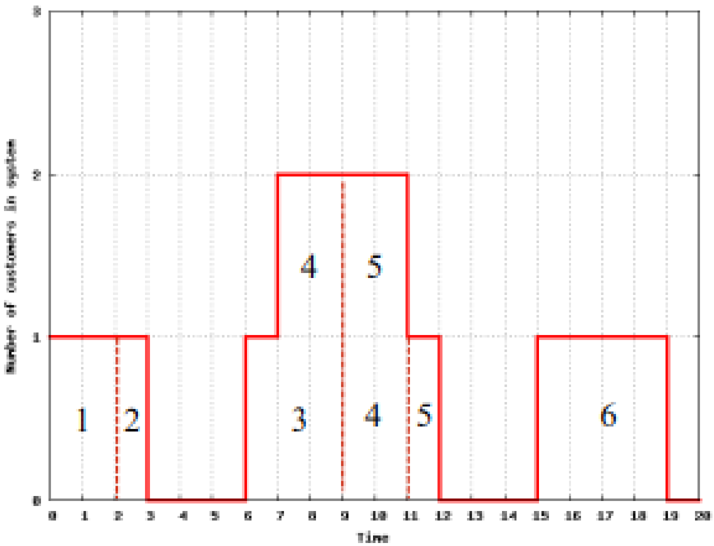
Chronological ordering of events

Clock Time	Customer Number	Event Type	Number of customers
0	1	Arrival	1
2	1	Departure	0
2	2	Arrival	1
3	2	Departure	0
6	3	Arrival	1
7	4	Arrival	2
9	3	Departure	1
9	5	Arrival	2
11	4	Departure	1
12	5	Departure	0
15	6	Arrival	1
19	6	Departure	0

Interesting observations

- Customer 1 is in the system at time 0
- Sometimes, there are no customers
- Sometimes, there are two customers
- Several events may occur at the same time

Number of customers in the system



SIM. A SINGLE-SERVER QUEUE

- **Example 1**
- **The Grocery Checkout, a Single-Server Queue:**

SIM. A SINGLE-SERVER QUEUE

- A small grocery store has one checkout counter.
- Customers arrive at the checkout counter at random times that range from 1 to 8 minutes apart.
- We assume that interarrival times are integer-valued with each of the 8 values having equal probability; this is a discrete uniform distribution.

Interarrival Time [minute]
1
2
3
4
5
6
7
8

SIM. A SINGLE-SERVER QUEUE

- A small grocery store has one checkout counter.
- Customers arrive at the checkout counter at random times that range from 1 to 8 minutes apart.
- We assume that interarrival times are integer-valued with each of the 8 values having equal probability; this is a discrete uniform distribution.

Interarrival Time [minute]	Probability
1	0.125
2	0.125
3	0.125
4	0.125
5	0.125
6	0.125
7	0.125
8	0.125

SIM. A SINGLE-SERVER QUEUE

- A small grocery store has one checkout counter.
- Customers arrive at the checkout counter at random times that range from 1 to 8 minutes apart.
- We assume that interarrival times are integer-valued with each of the 8 values having equal probability; this is a discrete uniform distribution.

Interarrival Time [minute]	Probability	Cumulative Probability
1	0.125	0.125
2	0.125	0.250
3	0.125	0.375
4	0.125	0.500
5	0.125	0.625
6	0.125	0.750
7	0.125	0.875
8	0.125	1.000

SIM. A SINGLE-SERVER QUEUE

- **Distribution of Time Between Arrivals**

Time between Arrivals (Minutes)	*Probability*	*Cumulative Probability*	*Random-Digit Assignment*
1	0.125	0.125	001–125
2	0.125	0.250	126–250
3	0.125	0.375	251–375
4	0.125	0.500	376–500
5	0.125	0.625	501–625
6	0.125	0.750	626–750
7	0.125	0.875	751–875
8	0.125	1.000	876–000

SIM. A SINGLE-SERVER QUEUE

- **Time-Between-Arrivals Determination**

Customer	Random Digits	Time between Arrivals (Minutes)	Customer	Random Digits	Time between Arrivals (Minutes)
1	–	–	11	109	1
2	913	8	12	093	1
3	727	6	13	607	5
4	015	1	14	738	6
5	948	8	15	359	3
6	309	3	16	888	8
7	922	8	17	106	1
8	753	7	18	212	2
9	235	2	19	493	4
10	302	3	20	535	5

SIM. A SINGLE-SERVER QUEUE

- The service times vary from 1 to 6 minutes (also integer valued), With The Probabilities Shown In The Following table.

Service Time [minute]	Probability	Cumulative Probability
1	0.10	0.10
2	0.20	0.30
3	0.30	0.60
4	0.25	0.85
5	0.10	0.95
6	0.05	1.00

SIM. A SINGLE-SERVER QUEUE

- **Service Time Distribution**

Service Time (Minutes)	Probability	Cumulative Probability	Random-Digit Assignment
1	0.10	0.10	01 – 10
2	0.20	0.30	11 – 30
3	0.30	0.60	31 – 60
4	0.25	0.85	61 – 85
5	0.10	0.95	86 – 95
6	0.05	1.00	96 – 00

SIM. A SINGLE-SERVER QUEUE

- **Service Times Generated**

Customer	Random Digits	Service Time (Minutes)	Customer	Random Digits	Service Time (Minutes)
1	84	4	11	32	3
2	10	1	12	94	5
3	74	4	13	79	4
4	53	3	14	05	1
5	17	2	15	79	5
6	79	4	16	84	4
7	91	5	17	52	3
8	67	4	18	55	3
9	89	5	19	30	2
10	38	3	20	50	3

SIM. A SINGLE-SERVER QUEUE

- **Simulation Table for Queueing Problem (20 customers)**

Simulation System						Performance Measure		
Customer	Interarrival Time [Minutes]	Arrival Time [Clock]	Service Time [Minutes]	Time Service Begins [Clock]	Time Service Ends [Clock]	Waiting Time in Queue [Minutes]	Time Customer in System [Minutes]	Idle Time of Server [Minutes]
1	—	0	4	0				
2	8		1					
3	6		4					
4	1		3					
5	8		2					
6	3		4					
7	8		5					
8	7		4					
9	2		5					
10	3		3					
11	1		3					
12	1		5					
13	5		4					
14	6		1					
15	3		5					
16	8		4					
17	1		3					
18	2		3					
19	4		2					
20	5		3					
Total	82		68					

SIM. A SINGLE-SERVER QUEUE

- **Simulation Table for Queueing Problem (20 customers)**

	Simulation System					Performance Measure		
Customer	Interarrival Time [Minutes]	Arrival Time [Clock]	Service Time [Minutes]	Time Service Begins [Clock]	Time Service Ends [Clock]	Waiting Time in Queue [Minutes]	Time Customer in System [Minutes]	Idle Time of Server [Minutes]
1	—	0	4	0	4	0	4	0
2	8	8	1					
3	6		4					
4	1		3					
5	8		2					
6	3		4					
7	8		5					
8	7		4					
9	2		5					
10	3		3					
11	1		3					
12	1		5					
13	5		4					
14	6		1					
15	3		5					
16	8		4					
17	1		3					
18	2		3					
19	4		2					
20	5		3					
Total	82		68					

SIM. A SINGLE-SERVER QUEUE

- **Simulation Table for Queueing Problem (20 customers)**

Simulation System | Performance Measure

Customer	Interarrival Time [Minutes]	Arrival Time [Clock]	Service Time [Minutes]	Time Service Begins [Clock]	Time Service Ends [Clock]	Waiting Time in Queue [Minutes]	Time Customer in System [Minutes]	Idle Time of Server [Minutes]
1	—	0	4	0	4	0	4	0
2	8	8	1	8	9	0	1	4
3	6		4					
4	1		3					
5	8		2					
6	3		4					
7	8		5					
8	7		4					
9	2		5					
10	3		3					
11	1		3					
12	1		5					
13	5		4					
14	6		1					
15	3		5					
16	8		4					
17	1		3					
18	2		3					
19	4		2					
20	5		3					
Total	82		68					

Simulation System						Performance Measure		
Customer	Interarrival Time [Minutes]	Arrival Time [Clock]	Service Time [Minutes]	Time Service Begins [Clock]	Time Service Ends [Clock]	Waiting Time in Queue [Minutes]	Time Customer in System [Minutes]	Idle Time of Server [Minutes]
1	–	0	4	0	4	0	4	0
2	8	8	1	8	9	0	1	4
3	6	14	4	14	18	0	4	5
4	1	15	3	18	21	3	6	0
5	8	23	2	23	25	0	2	2
6	3	26	4	26	30	0	4	1
7	8	34	5	34	39	0	5	4
8	7	41	4	41	45	0	4	2
9	2	43	5	45	50	2	7	0
10	3	46	3	50	53	4	7	0
11	1	47	3	53	56	6	9	0
12	1	48	5	56	61	8	13	0
13	5	53	4	61	65	8	12	0
14	6	59	1	65	66	6	7	0
15	3	62	5	66	71	4	9	0
16	8	70	4	71	75	1	5	0
17	1	71	3	75	78	4	7	0
18	2	73	3	78	81	5	8	0
19	4	77	2	81	83	4	6	0
20	5	82	3	83	86	1	4	0
Total	82		68			56	124	18

Simulation System						Performance Measure		
Customer	Interarrival Time [Minutes]	Arrival Time [Clock]	Service Time [Minutes]	Time Service Begins [Clock]	Time Service Ends [Clock]	Waiting Time in Queue [Minutes]	Time Customer in System [Minutes]	Idle Time of Server [Minutes]
1	—	0	4	0	4	0	4	0
2	8	8	1	8	9	0	1	4
3	6	14	4	14	18	0	4	5
4	1	15	3	18	21	3	6	0
5	8	23	2	23	25	0	2	2
6	3	26	4	26	30	0	4	1
7	8	34	5	34	39	0	5	4
8	7	41	4	41	45	0	[illegible]	2
9	2	43	5	45	50	[illegible]	[illegible]	[illegible]
10	3	46	3	50	53	[illegible]	[illegible]	[illegible]
11	1	47	3	53	56	[illegible]	[illegible]	[illegible]
12	1	48	5	56	61	[illegible]	[illegible]	[illegible]
13	5	53	4	61	65	[illegible]	[illegible]	[illegible]
14	6	59	1	65	66	[illegible]	[illegible]	[illegible]
15	3	62	5	66	71	4	9	0
16	8	70	4	71	75	1	5	0
17	1	71	3	75	78	4	7	0
18	2	73	3	78	81	5	8	0
19	4	77	2	81	83	4	6	0
20	5	82	3	83	86	1	4	0
Total	82		68			56	124	18

Total Run Time of Simulation

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\begin{array}{c}\text{average waiting time}\\ \text{(minutes)}\end{array} = \frac{\text{total time customers wait in queue (minutes)}}{\text{total numbers of customers}}$$

$$= \frac{56}{20} = 2.8 \text{ minutes}$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\text{probability (wait)} = \frac{\text{number of customers who wait}}{\text{total number of customers}}$$

$$= \frac{13}{20} = 0.65$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\text{probability of idle server} = \frac{\text{total idle time of server (minutes)}}{\text{total run time of simulation (minutes)}}$$

$$= \frac{18}{86} = 0.21$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\text{average service time (minutes)} = \frac{\text{total service time (minutes)}}{\text{total number of customers}}$$

$$= \frac{68}{20} = 3.4 \text{ minutes}$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\frac{\text{average time between}}{\text{arrivals (minutes)}} = \frac{\begin{array}{c}\text{sum of all times}\\ \text{between arrivals (minutes)}\end{array}}{\text{number of arrivals } -1}$$

$$= \frac{82}{19} = 4.3 \text{ minutes}$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\begin{array}{c}\text{Average waiting time of} \\ \text{those who wait (minutes)}\end{array} = \frac{\text{total time customers wait in queue (minutes)}}{\text{total number of customers who wait}}$$

$$= \frac{56}{13} = 4.3 \text{ minutes}$$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

$$\begin{array}{c}\text{average time customer}\\ \text{spends in the system}\\ \text{(minutes)}\end{array} = \frac{\begin{array}{c}\text{total time customers spend in the}\\ \text{system (minutes)}\end{array}}{\text{total number of customers}}$$

$$= \frac{124}{20} = 6.2 \text{ minutes}$$

SIM. A SINGLE-SERVER QUEUE

- Our objective is to analyze the system by simulating the arrival and service of 100 customers and to compute a variety of typical measures of performance for queueing models.
  - In actuality, 100 customers may be too small a sample size to draw reliable conclusions. Depending on our objectives, the accuracy of the results may be enhanced by increasing the sample size (number of customers), or by running multiple trials (or replications).

SIM. A SINGLE-SERVER QUEUE

- Our objective is to analyze the system by simulating the arrival and service of 100 customers and to compute a variety of typical measures of performance for queueing models.

  - A second issue is that of initial conditions. A simulation of a grocery store that starts with an empty system may or may not be realistic unless the intention is to model the system from startup or to model until steady-state operation is reached. Here, to keep calculations simple, the starting conditions are an empty grocery, and any concerns are overlooked.

SIMULATION TABLE FOR QUEUEING PROBLEM (100 CUSTOMERS)

Simulation System						Performance Measure		
Customer	Interarrival Time [Minutes]	Arrival Time [Clock]	Service Time [Minutes]	Time Service Begins [Clock]	Time Service Ends [Clock]	Waiting Time in Queue [Minutes]	Time Customer in System [Minutes]	Idle Time of Server [Minutes]
1	-	0	4	0	4	0	4	0
2	1	1	2	4	6	3	5	0
3	1	2	5	6	11	4	9	0
4	6	8	4	11	15	3	7	0
5	3	11	1	15	16	4	5	0
6	7	18	5	18	23	0	5	2
...								
100	5	415	2	416	418	1	3	0
Total	415		317			174	491	101

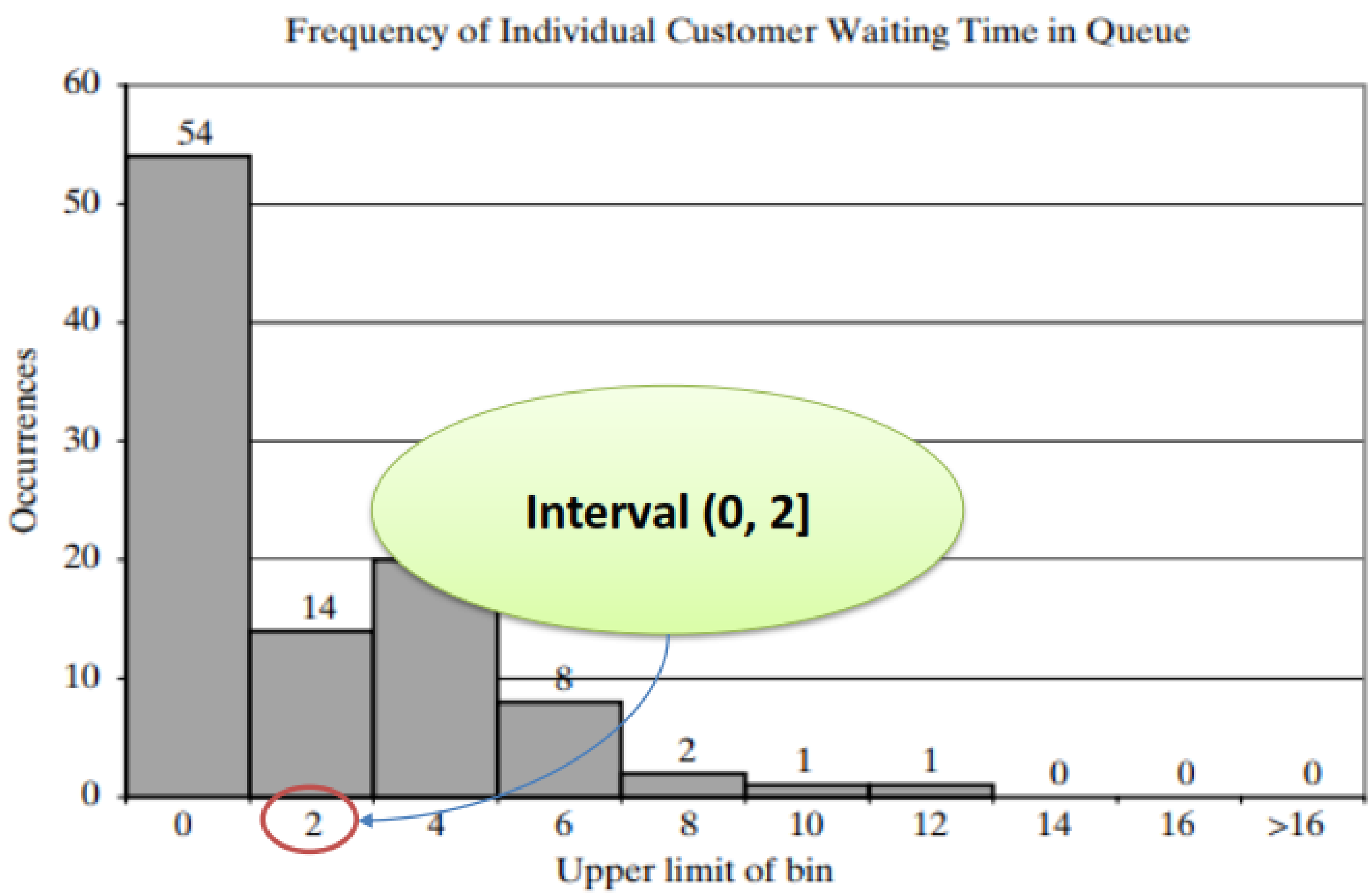
SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

Average waiting time	$\overline{w} = \frac{\sum \text{Waiting time in queue}}{\text{Number of customers}} = \frac{174}{100} = 1.74 \text{ min}$
Probability that a customer has to wait	$p(\text{wait}) = \frac{\text{Number of customer who wait}}{\text{Number of customers}} = \frac{46}{100} = 0.46$
Proportion of server idle time	$p(\text{idle server}) = \frac{\sum \text{Idle time of server}}{\text{Simulation run time}} = \frac{101}{418} = 0.24$
Average service time	$\bar{s} = \frac{\sum \text{Service time}}{\text{Number of customers}} = \frac{317}{100} = 3.17 \text{ min}$ $E(s) = \sum_{s=0}^{\infty} s \cdot p(s) = 0.1 \cdot 10 + 0.2 \cdot 20 + \cdots + 0.05 \cdot 6 = 3.2 \text{ min}$

SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

Average time between arrivals	$\bar{\lambda} = \dfrac{\sum \text{Times between arrivals}}{\text{Number of arrivals - 1}} = \dfrac{415}{99} = 4.19 \text{ min}$ $E(\lambda) = \dfrac{a+b}{2} = \dfrac{1+8}{2} = 4.5 \text{ min}$
Average waiting time of those who wait	$\bar{w}_{waited} = \dfrac{\sum \text{Waiting time in queue}}{\text{Number of customers that wait}} = \dfrac{174}{54} = 3.22 \text{ min}$
Average time a customer spends in system	$\bar{t} = \dfrac{\sum \text{Time customers spend in system}}{\text{Number of customers}} = \dfrac{491}{100} = 4.91 \text{ min}$ $\bar{t} = \bar{w} + \bar{s} = 1.74 + 3.17 = 4.91 \text{min}$

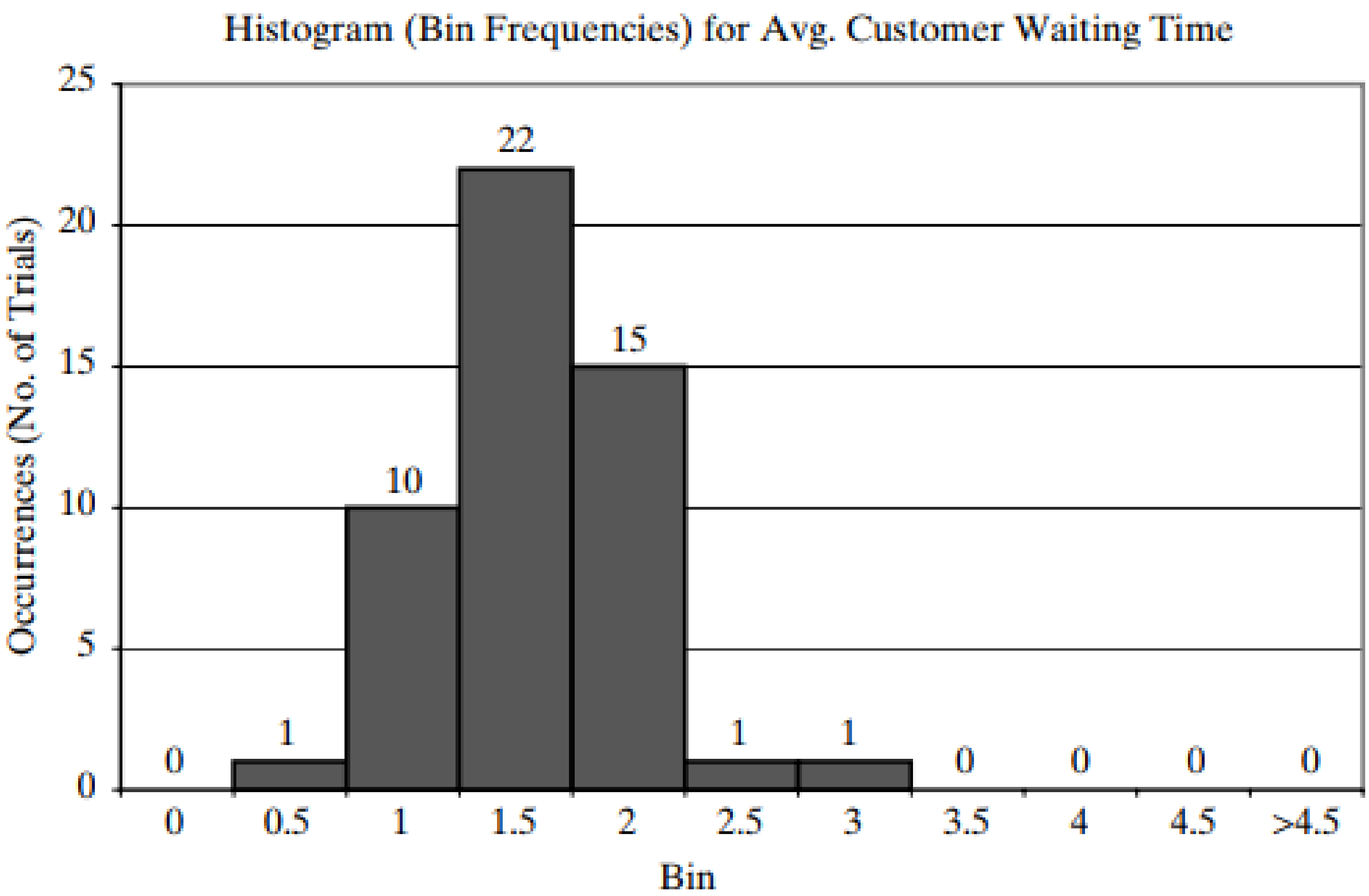
SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS



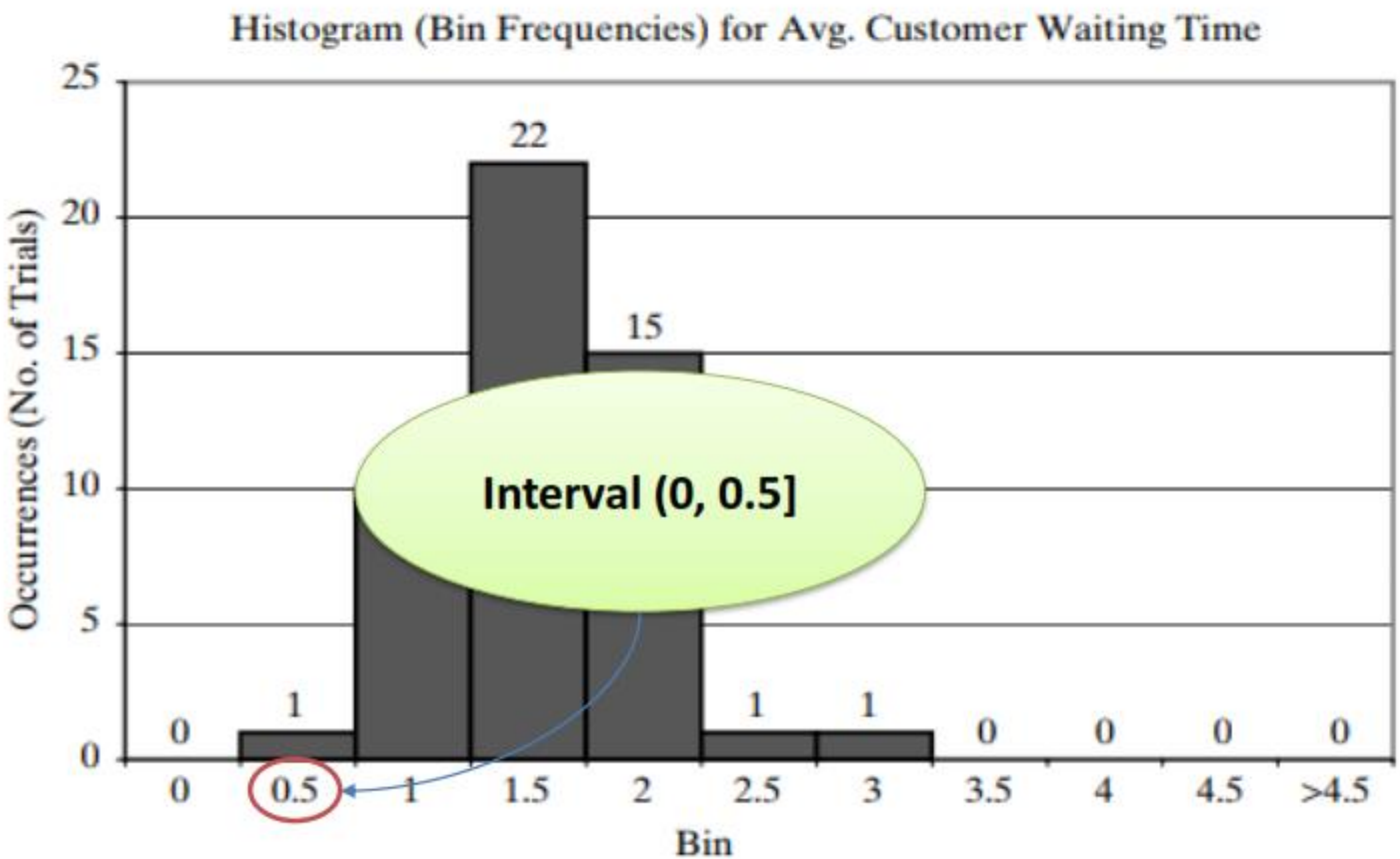
SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS

- We are now interested in running an experiment to address the question of how the average time in queue for the first 100 customers varies from day to day.
- Running once with 100 customers corresponds to one day. Running it 50 times corresponds to 50 days, where each trial represents one day.
- The overall average waiting time over 50 trials was 1.32 minutes.

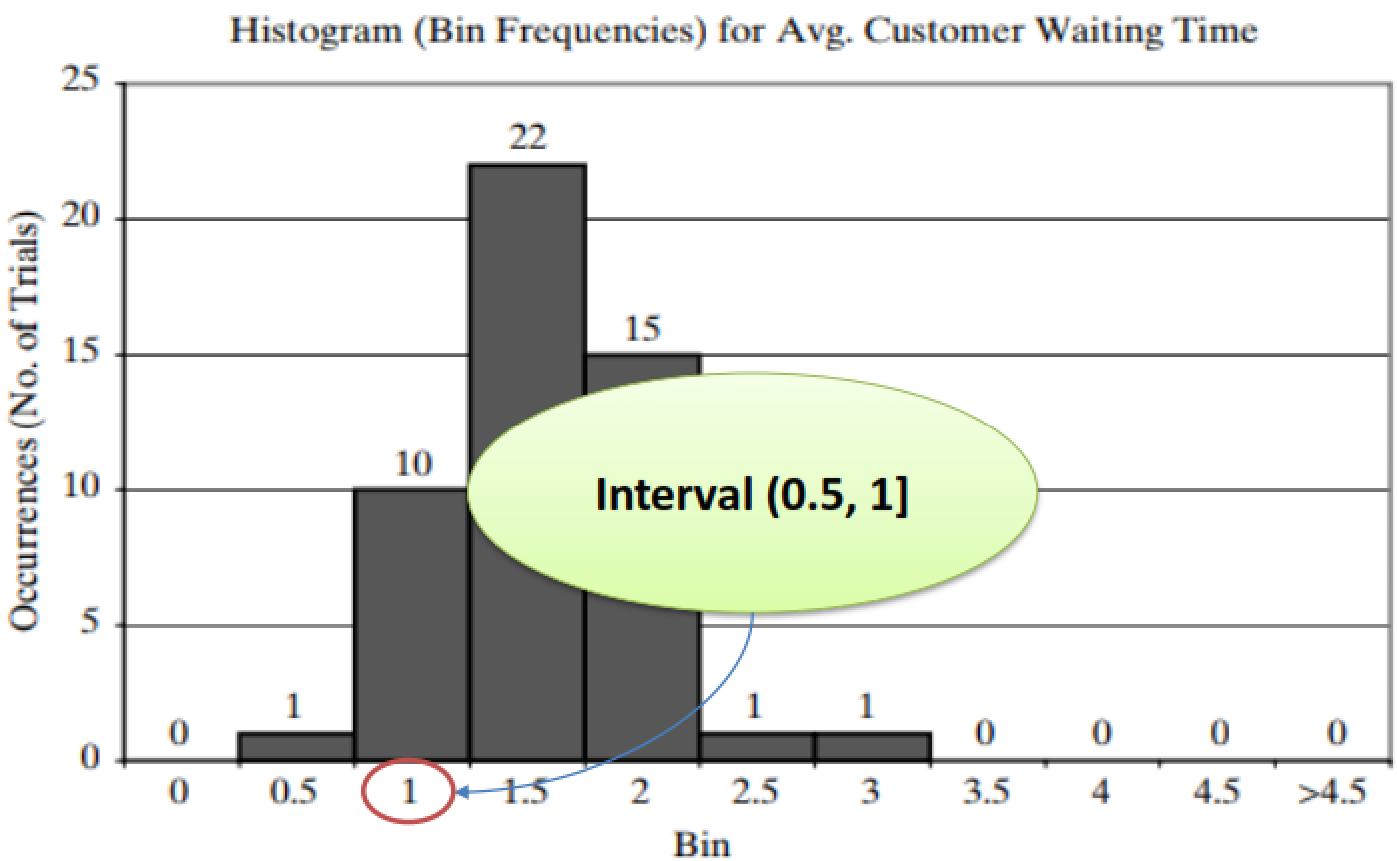
SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS



SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS



SOME OF THE FINDINGS FROM THE SIMULATION ARE AS FOLLOWS



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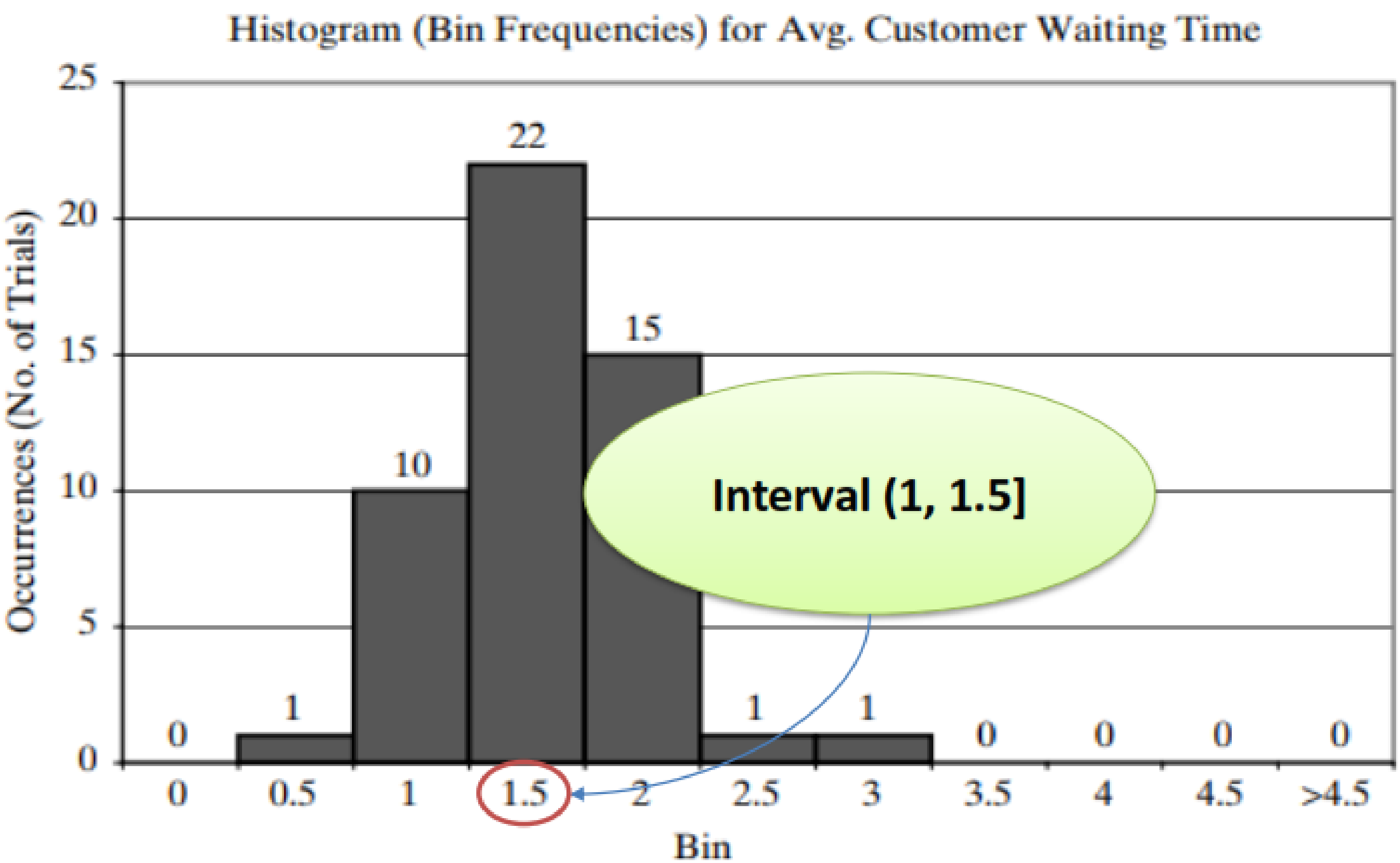

Exercise 1

- At a grocery store with one counter, customers arrive at random from 1 to 10 minutes apart (inter-arrival times have the same probability of occurrence). The service times vary from 1 to 8 minutes with the probabilities as 0.1, 0.18, 0.22, 0.16, 0.14, 0.10, 0.05 & 0.05 respectively. Analyze the system by simulating the arrival and service of 10 customers. Use the random numbers: 92, 73, 02, 95, 31, 92, 75, 24, 23, and 30 for inter-arrival time and the random numbers 84, 10, 74, 53, 17, 79, 91, 67, 89, 38 for the service time. Calculate for each customer the arrival time, service starting and ending times, waiting time in queue, time spent in the system, and idle time in the server. Show the results in the simulation table.

Thanks

Any questions